

null hypothesis of no effect. Compared to the Bayes factors in the RPCB, the evidence is stronger, possibly due to the mostly larger sample sizes in the RPP and EPRP.

Interestingly, the opposite conclusion is reached when we analyze the data using an equivalence test with a margin of $\Delta = 0.2$ (which may be considered liberal as it represents a small to medium effect based on the [Cohen, 1992](#) convention). In this case, equivalence at the 5% level can only be established for the [Ranganath and Nosek \(2008\)](#) original study and its replication simultaneously, as the confidence intervals from the other studies are too wide to be included in the equivalence range. Furthermore, the [Ranganath and Nosek \(2008\)](#) replication also illustrates the conceptual difference between testing for an *exactly zero* effect versus testing for an effect *within an interval around zero*. That is, the Bayes factor indicates no evidence for a zero effect (because the estimate is clearly not zero), but the equivalence test indicates evidence for a negligible effect (because the estimate is clearly within the equivalence range).

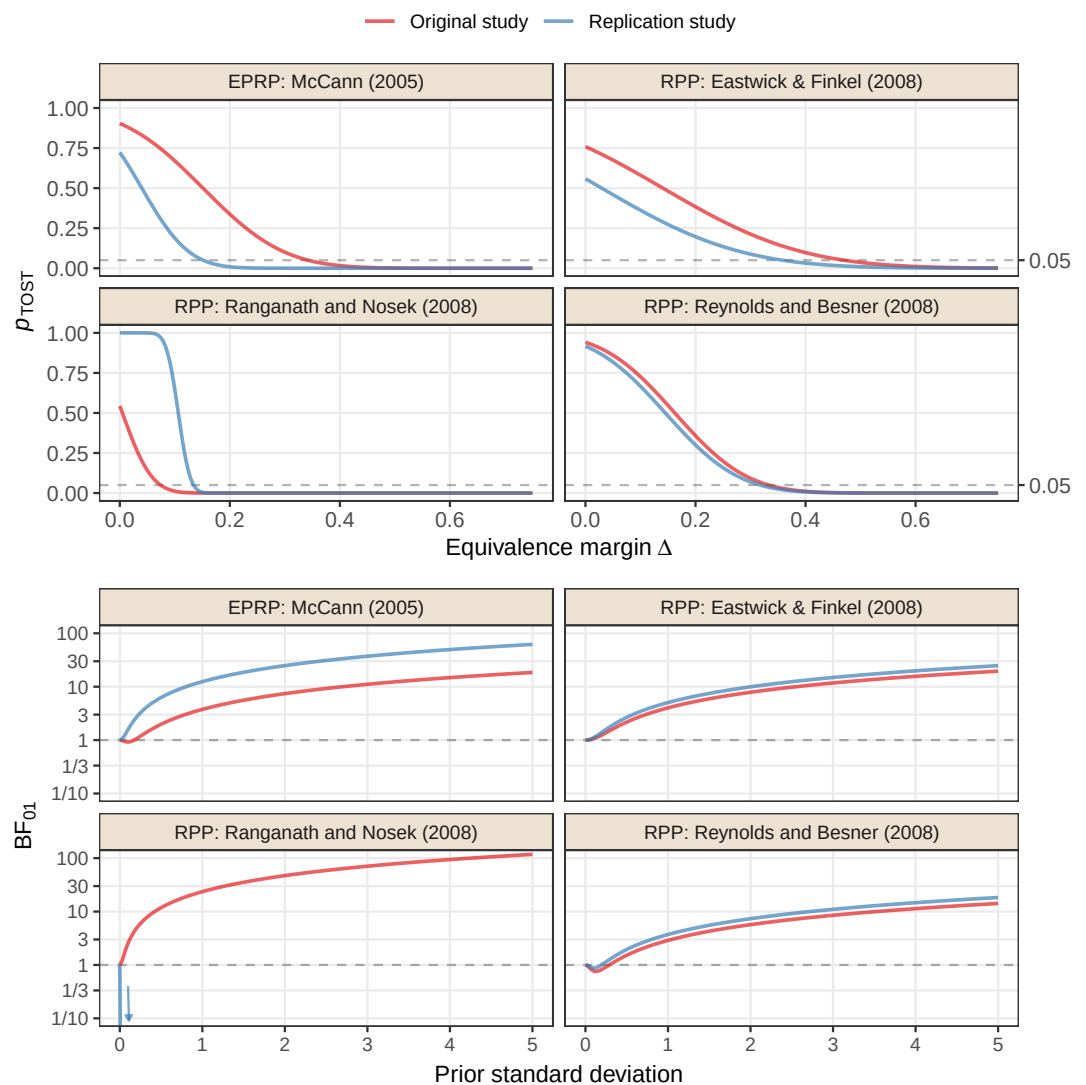


Figure 6. Sensitivity analyses for the “null results” and their replication studies from the Reproducibility Project: Psychology (RPP, [Open Science Collaboration, 2015](#)) and the Experimental Philosophy Replicability Project (EPRP, [Cova et al., 2018](#)). The Bayes factor of the replication of [Ranganath and Nosek \(2008\)](#) decreases very quickly and is only shown for a limited range.

As before, the particular choices of the equivalence margin Δ for the equivalence test and prior standard deviation of the Bayes factor are debatable. We therefore report sensitivity analyses in Figure 6 which show the TOST p -values and Bayes factors of original and replication studies for a range of margins and prior standard deviations, respectively. Apart from the **Ranganath and Nosek (2008)** study pair, all studies require large margins of about $\Delta = 0.4$ to establish replication success at the 5% level (in the sense of original and replication TOST p -values being smaller than 0.05). On the other hand, in all but the **Ranganath and Nosek (2008)** replication, the data provide substantial evidence for a null effect ($\text{BF}_{01} > 3$) for prior standard deviations of about one, while larger prior standard deviations of about three are required for the data to indicate strong evidence ($\text{BF}_{01} > 10$) for a null effect, whereas the data from the **Ranganath and Nosek (2008)** replication provide very strong evidence against a null effect for all prior standard deviations considered.

Appendix C: Technical details on Bayes factors

We assume that effect estimates are normally distributed around an unknown effect size θ with known variance equal to their squared standard error, i.e.,

$$\hat{\theta}_i | \theta \sim N(\theta, \sigma_i^2)$$

for original ($i = o$) and replication ($i = r$). This framework is similar to meta-analysis and can be applied to many types of effect sizes and data (**Spiegelhalter et al., 2004**, Section 2.4). We want to quantify the evidence for the null hypothesis that the effect size is equal to a null effect ($H_0 : \theta = \theta_0$, typically $\theta_0 = 0$) against the alternative hypothesis that the effect size is non-null ($H_1 : \theta \neq \theta_0$). This requires specification of a prior distribution for the effect size under the alternative, and we will assume a normal prior $\theta | H_1 \sim N(m, s^2)$ in the following. The Bayes factor based on an effect estimate is then given by the ratio of its likelihood under the null hypothesis to its marginal likelihood under the alternative hypothesis, i.e.,

$$\begin{aligned} \text{BF}_{01,i} &= \frac{p(\hat{\theta}_i | H_0)}{p(\hat{\theta}_i | H_1)} \\ &= \frac{p(\hat{\theta}_i | \theta_0)}{\int_{-\infty}^{+\infty} p(\hat{\theta}_i | \theta) p(\theta | H_1) d\theta} \\ &= \sqrt{1 + \frac{s^2}{\sigma_i^2}} \exp \left[-\frac{1}{2} \left\{ \frac{(\hat{\theta}_i - \theta_0)^2}{\sigma_i^2} - \frac{(\hat{\theta}_i - m)^2}{\sigma_i^2 + s^2} \right\} \right]. \end{aligned}$$

In the main analysis we used a normal unit-information prior, that is, a normal distribution centered around the value of no effect ($m = 0$) with a standard deviation s corresponding to the standard error of an SMD estimate based on one observation (**Kass and Wasserman, 1995**). Assuming that the group means are normally distributed $\bar{X}_1 \sim N(\theta_1, 2\tau^2/n)$ and $\bar{X}_2 \sim N(\theta_2, 2\tau^2/n)$ with n the total sample size and τ the known data standard deviation, the distribution of the SMD is $\text{SMD} = (\bar{X}_1 - \bar{X}_2)/\tau \sim N\{(\theta_1 - \theta_2)/\tau, \sigma^2 = 4/n\}$. The standard error σ of the SMD based on one unit ($n = 1$), is hence 2, meaning that the standard deviation of the unit-information prior is $s = 2$.

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